

CE710.01 High Performance Computing

Date: 26.12.2013, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

- (1) All questions are compulsory.
- (2) Indicate clearly, the option you attempt along with its respective question Number.
- (3) Figures to right indicate marks.
- (4) Rough work is done in the last page of main supplementary; don't write anything on the question paper.

SECTION-I

Q-1

- | | | |
|-----|---|----|
| (a) | How HPVM project is to deliver supercomputer performance? | 04 |
| (b) | What are the design constraints of distributed shared memory? Explain each in detail. | 04 |
| (c) | Differentiate Vector and Parallel Processing. | 03 |

Q-2 Answer the following questions.

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|-----|--|----|
| (a) | Explain Globus components resource management in detail. | 05 |
| (b) | Write a short note on PLUS architecture. | 04 |
| (c) | What are the traditional communication mechanisms for Clusters? What the advantages and disadvantages of each? | 03 |

OR

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|-----|---|----|
| (b) | What is supercomputer and how do they differ from a personal computer? | 04 |
| (c) | Why is fault tolerance vital to the success of grid computing? Justify with an example. | 03 |

Q-3 Answer the following questions. (Any Three) **12**

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|-----|---|--|
| (a) | Explain layered grid service protocols and their relationship with the internet service protocols. | |
| (b) | Differentiate between dedicated server process and shared server process with an example. | |
| (c) | List and explain the classification of cluster node into different layers (from viewpoint of a process P). | |
| (d) | List out type of grid. Explain any of them. | |

P.T.O

SECTION-II

Q-4 Answer the following questions.

- (a) Draw the Remote paging System Structure. 02
- (b) What is throughput? Wire equation for it. Explain parameter of it. 02
- (c) What are the features of HiPPI switches? 03
- (d) What do you mean by virtualization? Compare the HyperV and Xen virtualization softwares. 04

Q-5 Answer the following questions.

- (a) What is major problem associated in start and tree topology network? How Fat tree can solve that problem? 03
 - (b) Explain any four typical techniques to optimize communication (interprocess communication)? 04
 - (c) Explain Checkpoint based Rollback Recovery. 05
- OR**
- (c) List the name of Distributed Load Balancing algorithms. Explain any two. 05

Q-6 Answer the following questions.(Any Three)

- (a) With the help of a diagram, explain the working of loosely coupled multiprocessor systems. 12
- (b) In Cluster environment, Parallel I/O can be categorized either By Regular or Irregular Problems. Write down problems which come under both the category.
- (c) Explain Basic Component of LEGION.
- (d) How to configure HA (High Availability) using two Linux machines running a Heartbeat? Explain with proper example.